

Class Activity-1-MDP

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Course Code: MLA0305

Course Name: Reinforcement learning

R code:

```
#=====
# MARKOV DECISION PROCESS (MDP)
#=====

states <- c("S1", "S2", "S3")
actions <- c("A1", "A2")

cat("=====\n")
cat("    MARKOV DECISION PROCESS (MDP)\n")
cat("=====\n\n")
cat("States:\n")
print(states)
cat("\nNumber of States :", length(states), "\n\n")
cat("Actions:\n")
print(actions)
cat("\nNumber of Actions :", length(actions), "\n\n")
transition <- data.frame(
  From_State = c(
    "S1","S1","S1","S1",
    "S2","S2","S2","S2",
    "S3","S3","S3","S3"
  ),
  Action = c(
    "A1","A2","A1","A2",
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    "A1","A2","A1","A2",
    "A1","A2","A1","A2"
),
To_State = c(
    "S2","S2","S3","S3",
    "S1","S1","S3","S3",
    "S1","S1","S2","S2"
),
Probability = c(
    0.6,0.4,
    0.2,0.8,
    0.7,0.3,
    0.5,0.5,
    0.9,0.1,
    0.4,0.6
),
stringsAsFactors = FALSE
)
cat("=====\n")
cat("Transition Probabilities\n")
cat("=====\n")
print(transition)
cat("\nTotal Transition Entries :", nrow(transition), "\n\n")
rewards <- data.frame(
    From_State = c(
        "S1","S1","S1","S1",
        "S2","S2","S2","S2",

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    "S3","S3","S3","S3"),
Action = c(
    "A1","A2","A1","A2",
    "A1","A2","A1","A2",
    "A1","A2","A1","A2" ),
To_State = c(
    "S2","S2","S3","S3",
    "S1","S1","S3","S3",
    "S1","S1","S2","S2"),
Reward = c(
    5,10,
    -1,-5,
    3,7,
    2,1,
    4,6,
    0,-2),
stringsAsFactors = FALSE)

cat("=====\n")
cat("Rewards Associated with Each Transition\n")
cat("=====\n")
print(rewards)
cat("\nTotal Reward Entries :", nrow(rewards), "\n\n")
cat("=====\n")
cat("Transition and Reward Details\n")
cat("=====\n")
mdp <- merge(
    transition,

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rewards,
by = c("From_State", "Action", "To_State"))
print(mdp)
cat("\n=====\\n")
cat("MDP SUMMARY\\n")
cat("=====\\n")
cat("Number of States      :", length(states), "\\n")
cat("Number of Actions     :", length(actions), "\\n")
cat("Transition Probability Rows :", nrow(transition), "\\n")
cat("Reward Entries        :", nrow(rewards), "\\n")
cat("\\nState Space : ")
cat(states, sep = " ")
cat("\\n\\nAction Space : ")
cat(actions, sep = " ")
cat("\\n\\nProgram Executed Successfully.\\n")
cat("=====\\n")
states <- c("S1", "S2", "S3")
P_A1 <- matrix(c(
  0.0, 0.6, 0.2, # From S1 -> S1,S2,S3
  0.7, 0.0, 0.5, # From S2 -> S1,S2,S3
  0.9, 0.4, 0.0 # From S3 -> S1,S2,S3
), nrow = 3, byrow = TRUE)
rownames(P_A1) <- states
colnames(P_A1) <- states
cat("=====\\n")
cat("Transition Probability Matrix for Action A1\\n")
cat("=====\\n")

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print(P_A1)
P_A2 <- matrix(c(
  0.0, 0.4, 0.8, # From S1 -> S1,S2,S3
  0.3, 0.0, 0.5, # From S2 -> S1,S2,S3
  0.1, 0.6, 0.0 # From S3 -> S1,S2,S3
), nrow = 3, byrow = TRUE)
rownames(P_A2) <- states
colnames(P_A2) <- states
cat("\n=====\\n")
cat("Transition Probability Matrix for Action A2\\n")
cat("=====\\n")
print(P_A2)
reward_matrix <- data.frame(
  Current_State = c(
    "S1","S1","S1","S1",
    "S2","S2","S2","S2",
    "S3","S3","S3","S3"),
  Action = c(
    "A1","A2","A1","A2",
    "A1","A2","A1","A2",
    "A1","A2","A1","A2"),
  Next_State = c(
    "S2","S2","S3","S3",
    "S1","S1","S3","S3",
    "S1","S1","S2","S2" ),
  Reward = c(
    5, 10,

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-1, -5,
3, 7,
2, 1,
4, 6,
0, -2),
stringsAsFactors = FALSE)
cat("=====\n")
cat("Reward Matrix\n")
cat("=====\n")
print(reward_matrix, row.names = FALSE)
cat("=====\n")
cat("EXPECTED IMMEDIATE REWARD CALCULATION\n")
cat("=====\n\n")
cat("Current State : S1\n\n")
# Action A1
cat("Action : A1\n")
cat("Calculation:\n")
cat("=  $(0.6 \times 5) + (0.2 \times -1)$ \n")
eir_s1_a1 <- 0.6*5 + 0.2*(-1)
cat("= ",0.6*5," + ",0.2*(-1)," \n")
cat("Expected Immediate Reward =", eir_s1_a1," \n\n")
# Action A2
cat("Action : A2\n")
cat("Calculation:\n")
cat("=  $(0.4 \times 10) + (0.8 \times -5)$ \n")
eir_s1_a2 <- 0.4*10 + 0.8*(-5)
cat("= ",0.4*10," + ",0.8*(-5)," \n")

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cat("Expected Immediate Reward =", eir_s1_a2,"\n\n")
cat("-----\n")
cat("Current State : S2\n\n")
# Action A1
cat("Action : A1\n")
cat("Calculation:\n")
cat("= (0.7 × 3) + (0.5 × 2)\n")
eir_s2_a1 <- 0.7*3 + 0.5*2
cat("= ",0.7*3," + ",0.5*2,"\n")
cat("Expected Immediate Reward =", eir_s2_a1,"\n\n")
# Action A2
cat("Action : A2\n")
cat("Calculation:\n")
cat("= (0.3 × 7) + (0.5 × 1)\n")
eir_s2_a2 <- 0.3*7 + 0.5*1
cat("= ",0.3*7," + ",0.5,"\n")
cat("Expected Immediate Reward =", eir_s2_a2,"\n\n")
cat("-----\n")
cat("Current State : S3\n\n")
# Action A1
cat("Action : A1\n")
cat("Calculation:\n")
cat("= (0.9 × 4) + (0.4 × 0)\n")
eir_s3_a1 <- 0.9*4 + 0.4*0
cat("= ",0.9*4," + ",0,"\n")
cat("Expected Immediate Reward =", eir_s3_a1,"\n\n")
# Action A2

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cat("Action : A2\n")
cat("Calculation:\n")
cat("= (0.1 × 6) + (0.6 × -2)\n")
eir_s3_a2 <- 0.1*6 + 0.6*(-2)
cat("= ",0.1*6," + ",0.6*(-2)," \n")
cat("Expected Immediate Reward =", eir_s3_a2," \n\n")
result <- data.frame(
  State = c("S1","S1","S2","S2","S3","S3"),
  Action = c("A1","A2","A1","A2","A1","A2"),
  Expected_Reward = c(
    eir_s1_a1,
    eir_s1_a2,
    eir_s2_a1,
    eir_s2_a2,
    eir_s3_a1,
    eir_s3_a2 ))
cat("=====\n")
cat("SUMMARY OF EXPECTED IMMEDIATE REWARDS\n")
cat("=====\n")
print(result, row.names = FALSE)
library(grid)
# Summary data
summary_table <- data.frame(
  State = c("S1","S1","S2","S2","S3","S3"),
  Action = c("A1","A2","A1","A2","A1","A2"),
  Transition = c("S1 → S2","S1 → S3",
    "S2 → S1","S2 → S3",

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      "S3 → S1","S3 → S2"),
Expected_Reward = c(2.8,0.0,3.1,2.6,3.6,-0.6),
stringsAsFactors = FALSE)
grid.newpage()
# Title
grid.text(
  "MARKOV DECISION PROCESS (MDP)\nExpected Immediate Reward
Summary",
  y = 0.95,
  gp = gpar(fontsize = 18, fontface = "bold", col = "navy"))
# Table position
start_x <- 0.08
start_y <- 0.83
cell_w <- 0.22
cell_h <- 0.09
headers <- c("Current State","Action","Transition","Expected Reward")
header_cols <- c("steelblue4","darkgreen","darkorange3","purple4")
for(i in 1:4){
  grid.rect(
    x = start_x + (i-1)*cell_w,
    y = start_y,
    width = cell_w,
    height = cell_h,
    just = "left",
    gp = gpar(fill = header_cols[i], col = "white", lwd = 2)
  )
  grid.text(
    headers[i],

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    x = start_x + (i-1)*cell_w + cell_w/2,
    y = start_y,
    gp = gpar(col = "white", fontsize = 12, fontface = "bold")
  )
}
state_col <- "lightcyan"
action_col <- "honeydew"
transition_col <- "moccasin"
reward_col <- "lavender"
for(r in 1:nrow(summary_table)){
  yy <- start_y - r*cell_h
  grid.rect(
    x = start_x,
    y = yy,
    width = cell_w,
    height = cell_h,
    just = "left",
    gp = gpar(fill = state_col, col = "black"))
  grid.text(summary_table$State[r],
    x = start_x + cell_w/2,
    y = yy,
    gp = gpar(fontsize = 12,fontface="bold"))
  grid.rect(
    x = start_x + cell_w,
    y = yy,
    width = cell_w,
    height = cell_h,

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just = "left",
gp = gpar(fill = action_col, col = "black"))
grid.text(summary_table$Action[r],
          x = start_x + 1.5*cell_w,
          y = yy,
          gp = gpar(fontsize = 12,fontface="bold"))
grid.rect(
  x = start_x + 2*cell_w,
  y = yy,
  width = cell_w,
  height = cell_h,
  just = "left",
  gp = gpar(fill = transition_col, col = "black"))
grid.text(summary_table$Transition[r],
          x = start_x + 2.5*cell_w,
          y = yy,
          gp = gpar(fontsize = 12))
grid.rect(
  x = start_x + 3*cell_w,
  y = yy,
  width = cell_w,
  height = cell_h,
  just = "left",
  gp = gpar(fill = reward_col, col = "black"))
)
grid.text(summary_table$Expected_Reward[r],
          x = start_x + 3.5*cell_w,

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        y = yy,
        gp = gpar(fontsize = 12,fontface="bold",col="blue"))
    }
grid.text("Legend",
        x = 0.12,
        y = 0.08,
        gp = gpar(fontsize=14,fontface="bold"))
grid.rect(x=0.28,y=0.08,width=0.03,height=0.03,
        gp=gpar(fill=state_col))
grid.text("State",x=0.34,y=0.08)
grid.rect(x=0.46,y=0.08,width=0.03,height=0.03,
        gp=gpar(fill=action_col))
grid.text("Action",x=0.53,y=0.08)
grid.rect(x=0.66,y=0.08,width=0.03,height=0.03,
        gp=gpar(fill=transition_col))
grid.text("Transition",x=0.76,y=0.08)
grid.rect(x=0.90,y=0.08,width=0.03,height=0.03,
        gp=gpar(fill=reward_col))
grid.text("Reward",x=0.97,y=0.08)
library(DiagrammeR)
grViz("
digraph MDP {
graph [
layout = dot,
rankdir = TB,
splines = ortho,
nodesep = 0.6,

```

```

ranksep = 0.8,
bgcolor = white]
node [
shape = rectangle,
style = 'rounded,filled',
fontname = Helvetica,
fontsize = 16,
color = black,
penwidth = 2]
A [label='MARKOV\nDECISION PROCESS\n(MDP)',
fillcolor='#1F77B4',
fontcolor='white',
width=2.5]
B [label='Define\nStates\n(S1, S2, S3)',
fillcolor='#2ECC71']
C [label='Define\nActions\n(A1, A2)',
fillcolor='#F4D03F']
D [label='Input\nTransition\nProbabilities',
fillcolor='#5DADE2']
E [label='Input\nReward\nMatrix',
fillcolor='#F1948A']
F [label='Expected\nImmediate\nReward',
fillcolor='#AF7AC5',
fontcolor='white']
G [label='Compare\nActions',
fillcolor='#58D68D']
H [label='Select\nOptimal Policy',

```

```
fillcolor='#EB984E']
```

```
I [label='Final Output\nPolicy & Results',
```

```
fillcolor='#85C1E9']
```

```
A -> B
```

```
A -> C
```

```
B -> D
```

```
C -> D
```

```
D -> E
```

```
E -> F
```

```
F -> G
```

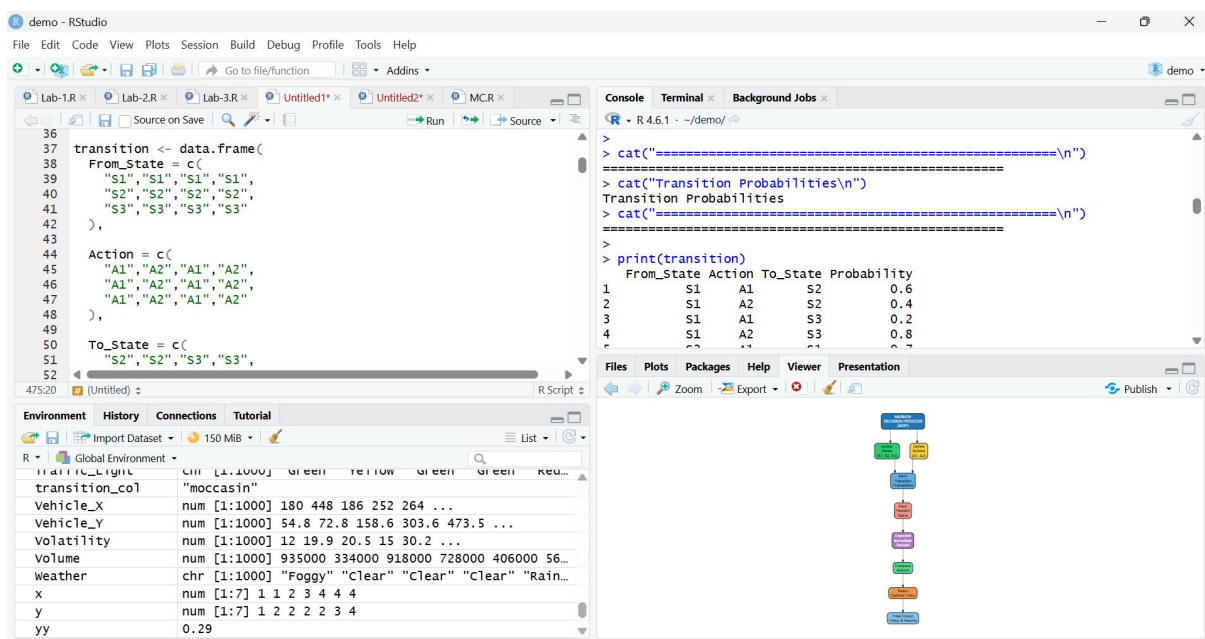
```
G -> H
```

```
H -> I
```

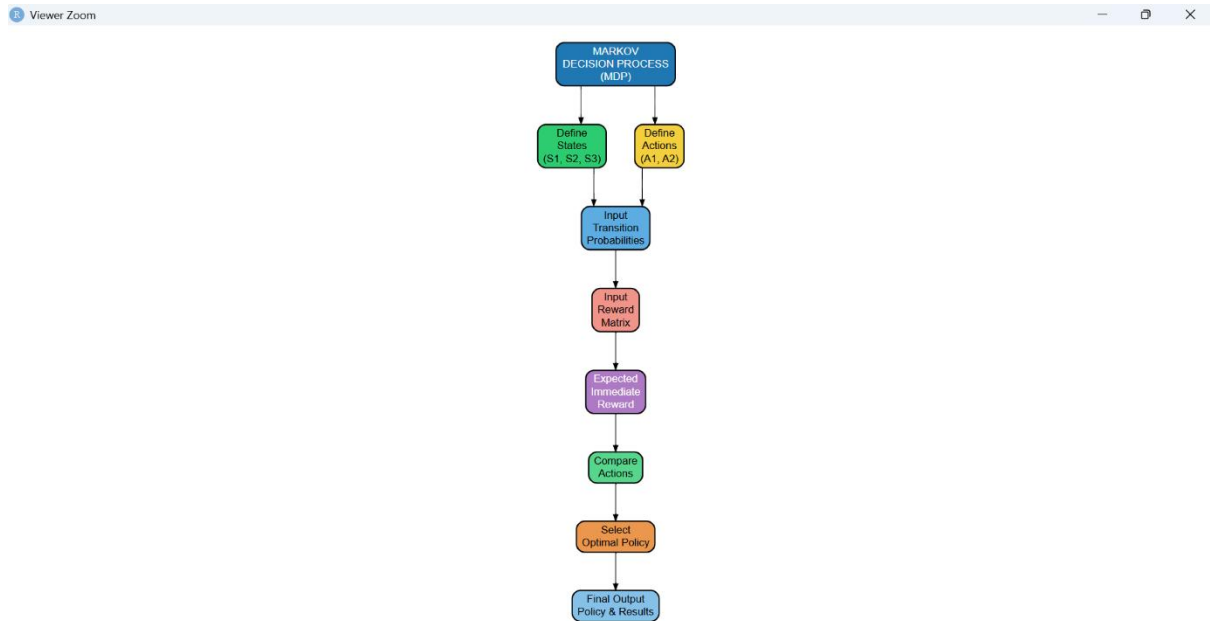
```
{rank=same; B C}
```

```
})
```

Full screenshot of Rstudio:



Visual result:



Summary result:

demo - RStudio

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```
> cat("=====\n")
> cat("Transition Probabilities\n")
Transition Probabilities
> cat("=====\n")
> print(transition)
  From_State Action To_State Probability
1      S1     A1    S2           0.6
2      S1     A2    S2           0.4
3      S1     A1    S3           0.2
4      S1     A2    S3           0.8
5      S2     A1    S1           0.7
6      S2     A2    S1           0.3
7      S2     A1    S3           0.5
8      S2     A2    S3           0.5
9      S3     A1    S1           0.9
10     S3     A2    S1           0.1
11     S3     A1    S2           0.4
12     S3     A2    S2           0.6
> cat("\nTotal Transition Entries :", nrow(transition), "\n\n")
Total Transition Entries : 12
>
> #=====
> # Reward Table
> #=====
```